

MIT 212 — Bilingual Question Bank

Weeks 1–2: MCQ • True/False • Short Answer • Code Output • Debugging • Practical | ද්විභාෂා ප්‍රශ්න

බැංකුව

MIT 212 — Programming Techniques • Master of Information Technology • Sabaragamuwa University of Sri Lanka

11. Multiple-Choice Questions (20) | බහුවරණ ප්‍රශ්න

Choose the best answer. Answers are given below each question for self-study; cover them while practising.

හොඳම පිළිතුර තෝරන්න. ස්වයං-අධ්‍යයනය සඳහා පිළිතුරු පහළින් දී ඇත; පුහුණු වන විට ඒවා වසන්න.

Q1. Programming is best described as:

ක්‍රමලේඛනය වඩාත් හොඳින් විස්තර වන්නේ:

- A) Drawing diagrams of computers
- B) Giving precise, step-by-step instructions to a computer to solve a problem
- C) Repairing computer hardware
- D) Designing web graphics

Answer: B

Q2. An algorithm is:

ඇල්ගොරිතමයක් යනු:

- A) A programming language
- B) A type of computer
- C) A step-by-step recipe for solving a task
- D) An error in a program

Answer: C

Q3. Which is language independent?

භාෂාවෙන් ස්වාධීන වන්නේ කුමක්ද?

- A) A Python program
- B) An algorithm
- C) An interpreter
- D) A compiler

Answer: B

Q4. A compiler:

සම්පාදකයක්:

- A) Translates and runs code line-by-line
- B) Translates the entire program first, then runs it
- C) Only finds logical errors
- D) Is used by Python

Answer: B

Q5. Python uses:

Python භාවිතා කරන්නේ:

- A) A compiler
- B) An interpreter
- C) Both at the same time
- D) Neither

Answer: B

Q6. Which environment lets you run Python in the browser using Google's servers for free?

Google සේවාදායක නොමිලේ භාවිතයෙන් බ්‍රවුසරයේ Python ධාවනය කළ හැක්කේ:

- A) MS Word
- B) Google Colab
- C) Photoshop

D) Windows Explorer

Answer: B

Q7. What is the output of `print("5 + 3")`?

`print("5 + 3")` හි ප්‍රතිදානය:

- A) 8
- B) 5 + 3
- C) 53
- D) Error

Answer: B — with quotes, Python treats it as literal text

Q8. What is the output of `print(5 + 3)`?

`print(5 + 3)` හි ප්‍රතිදානය:

- A) 8
- B) 5 + 3
- C) "8"
- D) Error

Answer: A — without quotes, Python performs the calculation

Q9. `print("Hello World)` causes:

`print("Hello World)` මගින් ඇතිවන්නේ:

- A) A logical error
- B) A syntax error — missing the closing quote
- C) Output: Hello World
- D) An infinite loop

Answer: B

Q10. Which statement about Python is TRUE?

Python ගැන සත්‍ය ප්‍රකාශය:

- A) Print and print are the same
- B) Python ignores spelling mistakes
- C) Python is case-sensitive
- D) Quotes are optional for text

Answer: C

Q11. Control flow allows us to:

පාලන ප්‍රවාහය මගින් හැකි වන්නේ:

- A) Change execution order based on conditions, or repeat steps
- B) Change the colour of code
- C) Make code run without a computer
- D) Delete variables

Answer: A

Q12. `elif` is short for:

`elif` යනු:

- A) "end life"
- B) "else if"
- C) "electric if"
- D) "element of if"

Answer: B

Q13. In an `if-elif-else` chain, the `else` block runs when:

`else` කොටස ක්‍රියාත්මක වන්නේ:

- A) The `if` is True
- B) Any `elif` is True
- C) None of the above conditions were True
- D) Always

Answer: C

Q14. `for i in range(5): print(i)` outputs:

ප්‍රතිදානය:

- A) 1 2 3 4 5
- B) 0 1 2 3 4
- C) 0 1 2 3 4 5
- D) 5 4 3 2 1

Answer: B — range stops before 5

Q15. A for loop is best when:

for ලූපය වඩාත් සුදුසු වන්නේ:

- A) The number of steps is unknown
- B) The number of steps is known
- C) You never want the loop to end
- D) There are no sequences

Answer: B

Q16. An infinite loop happens when:

අනන්ත ලූපයක් ඇතිවන්නේ:

- A) A for loop uses range()
- B) A while condition never becomes False
- C) A function has a return
- D) You use elif

Answer: B

Q17. The keyword used to create a function is:

ශ්‍රිතයක් සෑදීමේ මූල පදය:

- A) function
- B) create
- C) def
- D) fun

Answer: C

Q18. In def add_numbers(a, b): the names a and b are:

def add_numbers(a, b): හි a, b යනු:

- A) Arguments
- B) Parameters
- C) Return values
- D) Strings

Answer: B

Q19. return differs from print because return:

return, print ට වඩා වෙනස් වන්නේ:

- A) Shows text on the screen
- B) Sends the result back to where the function was called
- C) Deletes the function
- D) Only works in loops

Answer: B

Q20. Which is NOT a stated benefit of functions?

ශ්‍රිතවල වාසියක් නොවන්නේ:

- A) Modularity
- B) Reusability
- C) Readability
- D) Making programs run without errors

Answer: D

12. True/False Questions with Corrections (10) | සත්‍ය/අසත්‍ය ප්‍රශ්න සහ නිවැරදි කිරීම්

Q1. An algorithm can be written in English, Sinhala or as a flowchart.

ඇල්ගොරිතමයක් ඉංග්‍රීසි, සිංහල හෝ ගැලීම් සටහනකින් ලිවිය හැක.

Answer: TRUE.

Q2. Python uses a compiler.

Python සම්පාදකයක් භාවිතා කරයි.

Answer: FALSE — Python uses an INTERPRETER, which translates and runs code line-by-line.

Q3. A compiler translates a program line-by-line while it runs.

සම්පාදකය ධාවනය අතරතුර පේළියෙන් පේළිය පරිවර්තනය කරයි.

Answer: FALSE — a compiler translates the ENTIRE program first, then runs it; an interpreter works line-by-line.

Q4. `print(5 + 3)` displays 5 + 3 on the screen.

`print(5 + 3)` මගින් 5 + 3 ලෙස පෙන්වයි.

Answer: FALSE — without quotes Python performs the calculation, so it displays 8.

Q5. `Print("Hi")` and `print("Hi")` do the same thing.

`Print("Hi")` සහ `print("Hi")` සමානයි.

Answer: FALSE — Python is case-sensitive; `Print` with a capital P is not defined and causes an error.

Q6. The `else` block runs only when the `if` condition is True.

`if` කොන්දේසිය True වූ විට පමණක් `else` ක්‍රියාත්මක වේ.

Answer: FALSE — `else` runs when NONE of the previous conditions were True.

Q7. `range(1, 6)` produces the numbers 1, 2, 3, 4, 5.

`range(1, 6)` මගින් 1–5 ලැබේ.

Answer: TRUE — the stop value 6 is excluded.

Q8. A `while` loop is best when the number of repetitions is known in advance.

පුනරාවර්තන ගණන කලින් දන්නා විට `while` ලූපය සුදුසුයි.

Answer: FALSE — a `FOR` loop suits a known number of steps; `while` suits an UNKNOWN number.

Q9. A function runs automatically as soon as it is defined with `def`.

`def` කළ විගස ශ්‍රිතය ස්වයංක්‍රීයව ධාවනය වේ.

Answer: FALSE — a function does nothing until it is CALLED.

Q10. A variable created inside a function can be used anywhere in the program.

ශ්‍රිතය තුළ සාදන විචල්‍යය ඕනෑම තැනක භාවිතා කළ හැක.

Answer: FALSE — variable scope: variables created inside a function stay inside the function.

13. Definitions and Short-Answer Questions (10) | නිර්වචන සහ කෙටි පිළිතුරු ප්‍රශ්න

Q1. Define programming.

ක්‍රමලේඛනය නිර්වචනය කරන්න.

Answer: The art of giving precise, step-by-step instructions to a computer to solve a problem (e.g. mobile apps, banking systems, data analysis).

Q2. Define an algorithm and give an everyday example.

ඇල්ගොරිතමය නිර්වචනය කර එදිනෙදා උදාහරණයක් දෙන්න.

Answer: A step-by-step recipe for solving a task. Example — making tea: boil water → add tea → sweeten → serve.

Q3. What does the print() function do?

print() ශ්‍රිතයේ කාර්යය කුමක්ද?

Answer: It displays whatever is inside the quotes (or the value of the expression) onto the screen.

Q4. What is control flow?

පාලන ප්‍රවාහය යනු කුමක්ද?

Answer: By default Python runs top-to-bottom; control flow lets us change this order based on conditions or repeat steps (selection and loops).

Q5. State the purpose of the elif keyword.

elif මූල පදයේ අරමුණ කුමක්ද?

Answer: Short for "else if" — it tests an additional condition, checked only if the previous condition was False.

Q6. What is an infinite loop and how is it caused?

අනන්ත ලූපය යනු කුමක්ද? ඇතිවන්නේ කෙසේද?

Answer: A while loop that never ends because its condition never becomes False — typically caused by forgetting to update the counter.

Q7. Explain the 'black box' metaphor for functions.

ශ්‍රිත සඳහා 'black box' උපමාව පැහැදිලි කරන්න.

Answer: A function is like a machine: you give it Inputs (parameters), it performs a Process, and it gives you an Output (return value).

Q8. What is the difference between defining and calling a function?

ශ්‍රිතයක් නිර්වචනය කිරීම සහ කැඳවීම අතර වෙනස?

Answer: def creates (names) the block of code; calling — writing name() — actually runs it. A function does nothing until it is called.

Q9. What does return do?

return කරන්නේ කුමක්ද?

Answer: It sends the result back to where the function was called (so it can be stored, e.g. total = add_numbers(10, 5)) and ends the function.

Q10. List three benefits of using functions.

ශ්‍රිත භාවිතයේ වාසි තුනක් ලියන්න.

Answer: Modularity (break large problems into small pieces), Reusability (write once, use many times), Readability (named blocks are easier for humans to understand).

14. Comparison Questions (5) | සංසන්දන ප්‍රශ්න

Q1. Compare an algorithm with a program (two differences).

ඇල්ගොරිතමය සහ වැඩසටහන සසඳන්න (වෙනස්කම් දෙකක්).

Answer: (1) An algorithm is the step-by-step logic; a program is that logic written in a specific language. (2) An algorithm is language independent (English, Sinhala, flowchart); a program is language specific and can be executed by a computer.

Q2. Compare a compiler with an interpreter, and state which one Python uses.

සම්පාදකය/පරිවර්තකය සසඳා Python භාවිතා කරන එක සඳහන් කරන්න.

Answer: A compiler translates the entire program first then runs it (like translating a whole book); an interpreter translates and runs line-by-line (like a live speech translator). Python uses an interpreter.

Q3. Compare a for loop with a while loop under: best use, logic and ease of use.

for/while ලූප: සුදුසු භාවිතය, තර්කනය, පහසුව යටතේ සසඳන්න.

Answer: For — best for a known number of steps; iterates over sequences; cleaner syntax, hard to mess up. While — best for an unknown number of steps; checks a condition; requires careful condition update or it becomes infinite.

Q4. Compare print() with return.

print() සහ return සසඳන්න.

Answer: print shows a value on the screen for the human and cannot be stored; return sends the value back to the caller so it can be stored and reused (total = add(10,5)), and it also ends the function.

Q5. Compare a parameter with an argument using add_numbers as the example.

add_numbers උදාහරණයෙන් parameter/argument සසඳන්න.

Answer: Parameters are the variables inside the brackets of the definition — a and b in def add_numbers(a, b). Arguments are the actual values passed at call time — 10 and 5 in add_numbers(10, 5).

15. Code-Output Questions (8) | කේත ප්‍රතිදාන ප්‍රශ්න

For each snippet, write the exact output.

සෑම කේත කොටසකටම නිවැරදි ප්‍රතිදානය ලියන්න.

Q1.

```
print("Welcome to MIT")
print("My name is Nimal")
print("I am learning Python")
```

Answer: Welcome to MIT / My name is Nimal / I am learning Python — each on its own line.

Q2.

```
print("5 + 3")
print(5 + 3)
```

Answer: First line: 5 + 3 (literal text). Second line: 8 (calculation).

Q3.

```
x = 10
if x > 5:
    if x < 15:
        print("A")
    else:
        print("B")
```

Answer: A — x is 10, so x > 5 is True and x < 15 is True. (This is the slide's Interactive Challenge.)

Q4.

```
score = 75
if score >= 80:
    print("Grade: A")
elif score >= 60:
    print("Grade: B")
else:
    print("Grade: C")
```

Answer: Grade: B — 75 fails the >= 80 test but passes >= 60.

Q5.

```
for i in range(5):
    print(i)
```

Answer: 0 1 2 3 4 (each on a new line) — range stops before 5.

Q6.

```
fruits = ["Apple", "Banana"]
for f in fruits:
    print(f)
```

Answer: Apple, then Banana — the loop takes each element in turn.

Q7.

```
count = 1
```

```
while count <= 3:  
    print("Counting:", count)  
    count += 1
```

Answer: Counting: 1 / Counting: 2 / Counting: 3 — the loop stops when count becomes 4.

Q8.

```
def add_numbers(a, b):  
    result = a + b  
    return result  
  
total = add_numbers(10, 5)  
print(total)
```

Answer: 15 — return sends 15 back, it is stored in total and printed.

16. Error-Identification and Debugging Questions (6) | දෝෂ හඳුනාගැනීම් සහ නිරාකරණ ප්‍රශ්න

Identify the error type (syntax / runtime / logical), explain it, and write the corrected code.

දෝෂ වර්ගය (වාග් රීති / ධාවන / තාර්කික) හඳුනාගෙන, පැහැදිලි කර, නිවැරදි කේතය ලියන්න.

Q1.

```
print("Hello World)
```

Answer: Syntax error — the closing quote is missing. Python stops and points at the line. Fix: `print("Hello World")`

Q2.

```
Print("Welcome")
```

Answer: Syntax/name error — Python is case-sensitive; Print is not defined. Fix: `print("Welcome")`

Q3.

```
age = 20
if age >= 18
    print("Adult")
```

Answer: Syntax error — missing colon after the condition. Fix: `if age >= 18:`

Q4.

```
count = 1
while count <= 5:
    print(count)
```

Answer: Logical error (infinite loop) — the counter is never updated, so the condition never becomes False. Fix: add `count += 1` inside the loop.

Q5.

```
marks = input("Marks: ")
if marks >= 50:
    print("Pass")
```

Answer: Runtime error — `input()` returns a string, and a string cannot be compared with the number 50. Fix: `marks = int(input("Marks: "))`

Q6.

```
def area(w, h):
    print(w * h)

a = area(4, 5)
print(a + 10)
```

Answer: Logical/runtime error — the function prints instead of returning, so `a` is `None` and `a + 10` fails. Fix: use `return w * h`.

17. Algorithm / Pseudocode Questions (3) | ඇල්ගොරිතම / ව්‍යාජ කේත ප්‍රශ්න

Q1. Write an algorithm (plain steps) for making tea.

තේ සෑදීම සඳහා ඇල්ගොරිතමයක් ලියන්න.

```
Step 1: Boil water to 100°C.  
Step 2: Add tea leaves or a tea bag.  
Step 3: Sweeten – add sugar/milk.  
Step 4: Serve and enjoy the drink.
```

Q2. Write pseudocode for a grade calculator: read marks (0–100); 75+ → "Distinction", 50–74 → "Pass", below 50 → "Fail".

ලකුණු කියවා 75+ → Distinction, 50–74 → Pass, 50ට අඩු → Fail ලෙස ව්‍යාජ කේතය ලියන්න.

```
START  
READ marks  
IF marks >= 75 THEN PRINT "Distinction"  
ELSE IF marks >= 50 THEN PRINT "Pass"  
ELSE PRINT "Fail"  
END
```

Q3. Write pseudocode to find the sum of all even numbers between 1 and 20.

1–20 අතර ඉරට්ටේ සංඛ්‍යා එකතුව සෙවීමට ව්‍යාජ කේතය ලියන්න.

```
START  
SET total = 0  
FOR i FROM 1 TO 20  
    IF i MOD 2 = 0 THEN total = total + i  
END FOR  
PRINT total    (answer: 110)  
END
```

18. Flowchart Questions (2) | ගැලීම් සටහන් ප්‍රශ්න

Q1. Name the standard flowchart symbols for: start/end, a process, a decision, and flow direction, and state what a decision symbol must contain.

ආරම්භය/අවසානය, ක්‍රියාවලිය, තීරණය සහ ගැලීම් දිශාව සඳහා සම්මත සංකේත නම් කර, තීරණ සංකේතයේ අඩංගු විය යුත්තේ සඳහන් කරන්න.

Answer: Start/End = oval (terminator); Process = rectangle; Decision = diamond; Flow = arrows. A decision diamond must contain a question with True/False (Yes/No) exits — e.g. "input received?" with yes/no branches, as in the Week 2 slide.

Q2. Describe (or sketch) a flowchart for the input-validation loop shown in Week 2: read a value, repeat until input is received, exit when $a < 0$.

සතිය 2 හි ආදාන වලංගුකරණ ලූපයේ ගැලීම් සටහන විස්තර කරන්න: අගයක් කියවා, ආදානය ලැබෙන තුරු නැවත කර, $a < 0$ වූ විට ඉවත් වන්න.

Answer: Start → [a = input()] → <input received?> — No loops back to input; Yes → <is a < 0?> — No loops back to the start (read again); Yes → Exit. The two diamonds are decisions; arrows labelled yes/no show the repetition — this is exactly a while-loop structure.

19. Short Programming Questions (5) | කෙටි ක්‍රමලේඛන ප්‍රශ්න

Q1. Print your full name and age on two lines (Week 1, Task 1).

ඔබේ නම සහ වයස පේළි දෙකකින් මුද්‍රණය කරන්න.

Model answer:

```
print("Lasa Perera")
print("Age: 25")
```

Q2. Print this exact header using multiple print statements (the "MIT" Challenge).

"MIT" අභියෝගයේ ශීර්ෂය print ප්‍රකාශ කිහිපයකින් මුද්‍රණය කරන්න.

Model answer:

```
print("-----")
print("WELCOME TO MIT PROGRAM")
print("Faculty of Graduate Studies")
print("Sabaragamuwa University")
print("-----")
```

Q3. Grade calculator: ask for marks (0–100); print Distinction (75+), Pass (50–74) or Fail (below 50).

ලකුණු අසා Distinction / Pass / Fail මුද්‍රණය කරන්න.

Model answer:

```
marks = int(input("Enter marks (0-100): "))
if marks >= 75:
    print("Distinction")
elif marks >= 50:
    print("Pass")
else:
    print("Fail")
```

Q4. Use a loop to find the sum of all even numbers between 1 and 20 (hint: $i \% 2 == 0$).

1–20 අතර ඉරට්ටේ සංඛ්‍යා එකතුව ලූපයකින් සොයන්න.

Model answer:

```
total = 0
for i in range(1, 21):
    if i % 2 == 0:
        total += i
print(total) # 110
```

Q5. FizzBuzz: print 1–15, but multiples of 3 → "Fizz", multiples of 5 → "Buzz", multiples of both → "FizzBuzz".

1–15 මුද්‍රණය කරන්න; 3 ගුණාකාර → Fizz, 5 → Buzz, දෙකම → FizzBuzz.

Model answer:

```
for i in range(1, 16):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

Note for FizzBuzz: the "multiples of both" check MUST come first — if you test % 3 first, 15 would print "Fizz" and never reach "FizzBuzz". Order of conditions is a classic exam trap.

FizzBuzz සටහන: "දෙකම" පරීක්ෂාව මුලින්ම විය යුතුයි — නැතිනම් 15 ට "FizzBuzz" කිසිදා නොලැබේ. කොන්දේසි අනුපිළිවෙල සම්භාව්‍ය විභාග උගුලකි.

20. Substantial Practical Programming Problem | විශාල ප්‍රායෝගික ක්‍රමලේඛන ගැටලුව

Mini ATM with Login and Menu (combines Weeks 1–2) | ලොගින් සහ මෙනුවක් සහිත කුඩා ATM

Task: Build a simple ATM program. (a) The user gets a maximum of 3 tries to enter the correct PIN "1234"; after 3 wrong tries print "Card blocked" and stop. (b) After a successful login, show a menu repeatedly: 1 = Check balance, 2 = Deposit, 3 = Withdraw (reject if amount > balance), 4 = Exit. (c) Use separate functions for deposit and withdraw with parameters and return values. Starting balance is 5000.

කාර්යය: සරල ATM වැඩසටහනක් සාදන්න. (අ) නිවැරදි PIN "1234" ඇතුළත් කිරීමට උපරිම වාර 3ක්; වාර 3ම වැරදිවූහොත් "Card blocked" මුද්‍රණය කර නවතින්න. (ආ) සාර්ථක ලොගින් වූ පසු මෙනුව නැවත නැවත පෙන්වන්න: 1 = ශේෂය, 2 = තැන්පතු, 3 = ආපසු ගැනීම (ශේෂයට වඩා නම් ප්‍රතික්ෂේප), 4 = ඉවත්වීම. (ඇ) deposit/withdraw සඳහා parameters සහ return සහිත වෙනම ශ්‍රිත භාවිතා කරන්න. ආරම්භක ශේෂය 5000.

Model solution:

```
def deposit(balance, amount):
    balance = balance + amount
    print("Deposited:", amount)
    return balance

def withdraw(balance, amount):
    if amount > balance:
        print("Insufficient funds!")
    else:
        balance = balance - amount
        print("Withdrawn:", amount)
    return balance

def atm_process():
    correct_pin = "1234"
    balance = 5000
    tries = 0

    # (a) Login loop: maximum 3 tries
    while tries < 3:
        pin = input("Enter PIN: ")
        if pin == correct_pin:
            print("Login Success!")
            break
        else:
            tries += 1
            print("Wrong PIN. Try again.")

    if tries == 3:
        print("Card blocked")
        return # stop the program

    # (b) Menu loop
    while True:
        print("1) Balance  2) Deposit  3) Withdraw  4) Exit")
        choice = input("Choose: ")
        if choice == "1":
            print("Balance:", balance)
        elif choice == "2":
            amount = int(input("Amount: "))
            balance = deposit(balance, amount)
        elif choice == "3":
```

```

        amount = int(input("Amount: "))
        balance = withdraw(balance, amount)
    elif choice == "4":
        print("Thank you!")
        break
    else:
        print("Invalid choice")

```

```
atm_process()
```

Why this earns full marks | සම්පූර්ණ ලකුණු ලැබෙන හේතු

- while + counter implements the 3-try password loop (Week 2 homework).
while + ගණකය මගින් වාර 3 මුරපද ලූපය ක්‍රියාත්මක වේ.
- break exits the login loop on success; return stops everything when the card is blocked.
සාර්ථක වූ විට break; කාඩ්පත අවහිර වූ විට return.
- if-elif-else handles the menu branches, including an else for invalid input.
if-elif-else මෙනු ශාඛා හසුරුවයි.
- deposit/withdraw show parameters, return values, and the scope rule — balance must be returned and reassigned because variables inside a function stay inside.
deposit/withdraw මගින් parameters, return සහ විෂය පථ නීතිය පෙන්වයි — balance ආපසු ලබාදී නැවත පැවරිය යුතුය.
- input() values are converted with int() before arithmetic.
ගණනයට පෙර input() අගයන් int() මගින් හරවයි.